

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460806.90159 +/- 0.00092 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.051
Transit depth (Rp/Rs)^2	2.68 +/- 1.67 [%]
Orbital Inclination (inc)	86.15 +/- 1.81
Airmass coefficient 1 (a1)	0.791 +/- 0.015
Airmass coefficient 2 (a2)	0.204 +/- 0.08
Scatter in the residuals of the lightcurve fit is	1.84 %
Best Comparison Star	#2 - [592, 217]
Optimal Aperture	2.87
Optimal Annulus	14.68
Transit Duration (day)	0.05 +/- 0.045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.164 ± 0.051 vs 0.1356 (deviation 0.56σ)
Duration fit vs published	0.05 d vs 0.1129 d (deviation 1.4σ)
Mid-time O–C	-2.4 min
Depth SNR	3.2
Residual scatter	1.84 %

Light curve

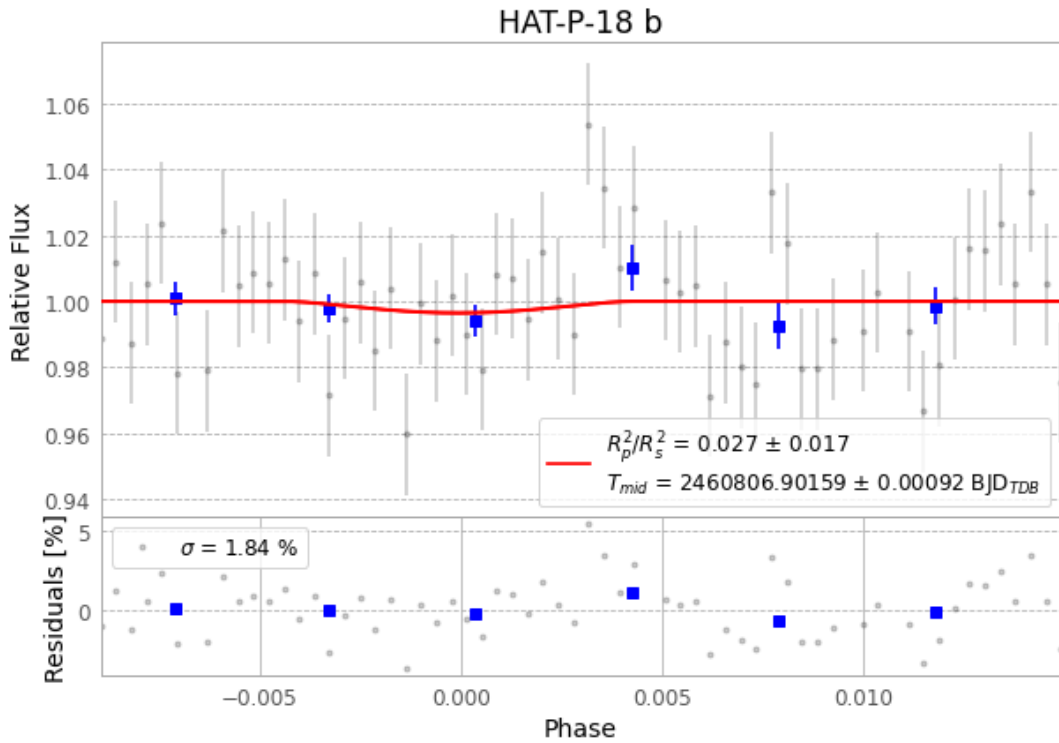


Figure 1 — FinalLightCurve_HAT-P-18 b_2025-05-11.png (SHA-256 d9dc1b2746834ce0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [320, 259], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 868963694443719f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), HATP-18250511082115.FITS → HATP-18250511113015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-250511.log (363af9392f43...), FinalLightCurve_HAT-P-18 b_2025-05-11.png (d9dc1b274683...), FinalLightCurve_HAT-P-18 b_2025-05-11.csv (7a86aff7cf70...)

Compute resources

Wall time 192.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 01:05Z.